

Course Title: Introduction to Artificial Intelligence Systems

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Course Description

This course introduces the fundamental concepts, techniques, and applications of **Artificial Intelligence (AI)**. It covers problem-solving methods, knowledge representation, machine learning basics, and real-world AI applications. Students will gain both theoretical understanding and practical exposure.

Course Objectives

By the end of this course, students should be able to:

- Understand the fundamentals of AI and its history
- Apply problem-solving techniques using AI approaches
- Represent knowledge using different models
- Understand basic machine learning concepts
- Develop simple AI-based applications

Course Outline (Weekly Breakdown)

Week 1: Introduction to AI

- Definition of AI
- History and evolution of AI
- AI applications in real life
- Types of AI (Narrow vs General AI)

Week 2: Intelligent Agents

- Concept of agents and environments
- Types of agents (simple reflex, model-based, goal-based, utility-based)
- Agent architectures

Week 3: Problem Solving in AI

- Problem formulation
- State space representation
- Search strategies overview

Week 7: Knowledge Representation

- Logic-based representation
- Propositional Logic
- First-Order Logic (FOL)

Week 8: Reasoning in AI

- Forward and backward chaining
- Inference mechanisms
- Rule-based systems

Week 9: Machine Learning Basics

- Introduction to Machine Learning
- Types: Supervised, Unsupervised, Reinforcement Learning
- Training vs Testing data

Week 10: Supervised Learning

- Classification and regression
- Algorithms: Decision Trees, k-NN
- Model evaluation

Week 11: Unsupervised Learning

- Clustering (K-means)
- Dimensionality reduction basics

Week 12: Natural Language Processing (NLP)

- Text processing basics
- Applications (chatbots, translation systems)

Week 13: Robotics and Computer Vision

- Introduction to robotics
- Image processing basics
- AI in vision systems

Week 14: Ethics and Future of AI

- Ethical issues in AI
- Bias and fairness

- Future trends in AI

Practical / Lab Work

- Implement search algorithms (BFS, DFS)
- Build a simple chatbot
- Basic machine learning model using Python (e.g., classification)
- Mini AI project (group work)

Assessment Methods

- Continuous Assessment (Assignments, Quizzes) – 30%
- Practical/Lab Work – 20%
- Midterm Exam – 20%
- Final Exam – 30%

Recommended Textbooks

- Stuart Russell & Peter Norvig – *Artificial Intelligence: A Modern Approach*
- Tom Mitchell – *Machine Learning*
- Elaine Rich – *Artificial Intelligence*

WEEK 1: INTRODUCTION TO ARTIFICIAL INTELLIGENCE

1.1 What is Artificial Intelligence?

Artificial Intelligence (AI) is the branch of computer science concerned with building systems that can perform tasks that normally require human intelligence.

Examples of Human Intelligence:

- Learning from experience
- Problem-solving
- Understanding language
- Recognizing patterns
- Decision-making

AI System Examples:

- Virtual assistants (Siri, Alexa)
- Recommendation systems (Netflix, YouTube)
- Self-driving cars
- Chatbots

1.2 Definitions of AI

Different experts define AI in different ways:

- “AI is the science and engineering of making intelligent machines.”
- “AI is the study of agents that perceive their environment and act upon it.”

AI can be viewed in 4 perspectives:

FOUR PERSPECTIVES OF ARTIFICIAL INTELLIGENCE

1. Thinking Humanly (The Cognitive Modeling Approach)

Definition

This approach focuses on building AI systems that think like humans—that is, they replicate human thought processes.

Key Idea

"How do humans think, and can we model that in a machine?"

How it works

- Studies human brain processes
- Uses insights from:
 - Psychology
 - Cognitive science
 - Neuroscience
- Attempts to simulate human reasoning step-by-step

Example

- A system that solves a problem the same way a human would (trial and error, reasoning, memory use)

Real-World Application

- Cognitive architectures like human-like learning systems
- Educational software that mimics student thinking patterns

Advantages

- Helps understand human intelligence
- Useful in human-computer interaction

Limitations

- Human thinking is complex and not fully understood
- Difficult to accurately model emotions, intuition, and creativity

2. Acting Humanly (The Turing Test Approach)

Definition

This approach focuses on building systems that behave like humans, regardless of how they think internally.

Key Idea

"Can a machine behave indistinguishably from a human?"

The Turing Test

Proposed by Alan Turing.

- A machine is considered intelligent if:
 - A human cannot distinguish it from another human through conversation

Requirements for Passing the Test:

1. Natural Language Processing (communication)
2. Knowledge Representation
3. Reasoning ability
4. Learning capability

Example

- Chatbots that mimic human conversation

Real-World Application

- Virtual assistants (e.g., Siri, Alexa)
- Customer service chatbots

Advantages

- Focuses on practical outcomes
- Easy to test (via interaction)

Limitations

- Does not measure *true intelligence*, only behavior
- A system may appear intelligent but not actually “understand”

3. Thinking Rationally (The Laws of Thought Approach)

Definition

This approach focuses on systems that think logically and correctly, based on formal reasoning. Key Idea

“Can a machine reason correctly using logic?”

How it works

- Uses formal logic (e.g., propositional logic, predicate logic)
- Follows strict rules of reasoning

Example

- If:
 - All humans are mortal
 - Socrates is a human
- Then:
 - Socrates is mortal

AI systems apply such logical rules automatically.

Real-World Application

- Expert systems
- Automated theorem proving
- Rule-based systems

Advantages

- Produces correct and consistent results
- Easy to verify correctness

Limitations

- Not all real-world problems can be expressed in logical form
- Can be computationally expensive
- Ignores uncertainty and incomplete information

4. Acting Rationally (The Rational Agent Approach)

Definition

This approach focuses on systems that act in the best possible way to achieve goals, based on available information.

Key Idea

“What is the best action to take?”

Rational Agent Concept

A rational agent:

- Perceives its environment
- Chooses actions that maximize success (performance measure)

Example

- A self-driving car choosing the safest and fastest route
- A chess AI selecting the best move

Real-World Application

- Robotics
- Autonomous systems
- Game-playing AI

Advantages

- Practical and widely used
- Handles uncertainty and dynamic environments
- Focuses on outcomes rather than internal processes

Limitations

- Requires accurate models of the environment
- May not always achieve perfect results due to incomplete information

1.3 History and Evolution of AI

KEY MILESTONES IN ARTIFICIAL INTELLIGENCE

1. 1950 – Alan Turing Proposes the Turing Test

What Happened

In 1950, Alan Turing published a groundbreaking paper titled *“Computing Machinery and Intelligence.”*

He introduced the idea of the Turing Test as a way to determine whether a machine can exhibit intelligent behavior.

The Turing Test Explained

- A human judge interacts with:
 - A human
 - A machine

- If the judge cannot reliably tell which is which, the machine is said to be intelligent.

Why It Matters

- First formal attempt to define machine intelligence
- Shifted focus from *“Can machines think?”* to *“Can machines behave intelligently?”*
- Inspired research in:
 - Natural Language Processing
 - Chatbots

2. 1956 – Dartmouth Conference (Birth of AI)

What Happened

- Held at Dartmouth College, USA
- Organized by researchers like:
 - John McCarthy
 - Marvin Minsky

This is where the term “Artificial Intelligence” was officially coined.

Key Goals

- Build machines that can:
 - Learn
 - Reason
 - Improve themselves

Why It Matters

- Marked the official beginning of AI as a field of study
- Attracted funding and global interest
- Led to early AI programs (problem-solving, symbolic reasoning)

3. 1970s–1980s – Rise of Expert Systems

What Happened

AI research shifted toward building Expert Systems—programs that mimic the decision-making of human experts.

What are Expert Systems?

- Systems that use:
 - Knowledge base (facts + rules)
 - Inference engine (decision-making logic)

Example

- Medical diagnosis systems (e.g., MYCIN)
- Industrial troubleshooting tools

Why It Matters

- First successful commercial AI applications
- Widely used in:
 - Healthcare
 - Finance
 - Engineering

Challenges

- Expensive to build and maintain
- Knowledge acquisition was difficult
- Led to a period called the “AI Winter” (reduced funding and interest)

4. 1997 – IBM's Deep Blue Defeats Garry Kasparov

What Happened

- Deep Blue, a supercomputer developed by IBM, defeated world chess champion Garry Kasparov.

How It Worked

- Evaluated millions of positions per second
- Used advanced search algorithms and heuristics

Why It Matters

- First time a computer defeated a reigning world chess champion
- Proved machines can outperform humans in complex intellectual tasks

Impact

- Boosted public confidence in AI
- Encouraged research in:
 - Game AI
 - Optimization problems

5. 2010s–Present – Growth of Machine Learning & Deep Learning

What Happened

AI experienced rapid growth due to:

- Availability of big data
- Increased computing power (GPUs)
- Advances in algorithms

Key Technologies

- Machine Learning (ML) → systems learn from data
- Deep Learning (DL) → neural networks with many layers

Applications

- Image recognition (face detection)
- Speech recognition
- Language translation
- Self-driving cars

Notable Systems

- AlphaGo defeating world champions in Go
- Virtual assistants and modern chatbots

Why It Matters

- AI became part of everyday life
- Major breakthroughs in:
 - Healthcare
 - Finance
 - Education

1.4 Applications of AI

AI is used in many fields:

Healthcare

- Disease diagnosis
- Medical imaging

Finance

- Fraud detection
- Algorithmic trading

Transportation

- Autonomous vehicles

Daily Life

- Voice assistants
- Face recognition

1.5 Types of AI

Based on Capability

Type	Description
Narrow AI	Performs specific tasks (e.g., chatbots)
General AI	Human-level intelligence (not yet achieved)
Super AI	Beyond human intelligence (theoretical)

1.6 AI System Structure

Basic AI Model:

Input → Processing → Output

- Input: Data (text, images, sound)
- Processing: Algorithms
- Output: Decisions/actions

AI System Overview

1.7 Advantages and Disadvantages

Advantages

- Reduces human effort
- High accuracy
- Works 24/7

Disadvantages

- Job displacement
- High cost
- Ethical concerns

1.8 ETHICAL ISSUES IN ARTIFICIAL INTELLIGENCE

As AI systems become more powerful and widely used, they raise important ethical concerns about fairness, safety, and responsibility. These issues must be carefully addressed to ensure AI benefits society.

1. Bias in Decision-Making

What is Bias in AI?

Bias occurs when an AI system produces unfair or discriminatory results due to flawed data or algorithms.

Causes of Bias

- Biased training data (e.g., incomplete or unbalanced datasets)
- Human prejudice embedded in system design
- Lack of diversity in development teams

Examples

- Facial recognition systems failing to recognize certain ethnic groups accurately
- Hiring algorithms favoring certain genders or backgrounds

Impact

- Discrimination in:
 - Employment
 - Lending/finance

- Criminal justice systems

Possible Solutions

- Use diverse and representative datasets
- Regular auditing of AI systems
- Implement fairness-aware algorithms

2. Privacy Concerns

What is Privacy in AI?

AI systems often require large amounts of personal data, raising concerns about how that data is collected, stored, and used.

Key Issues

- Unauthorized data collection
- Surveillance without consent
- Data breaches and misuse

Examples

- Social media platforms tracking user behavior
- AI-powered surveillance cameras monitoring individuals

Impact

- Loss of personal freedom
- Identity theft
- Misuse of sensitive information

Possible Solutions

- Strong data protection laws
- Data encryption and anonymization
- User consent and transparency

3. Autonomous Weapons

What are Autonomous Weapons?

Weapons that can select and attack targets without human intervention using AI.

Examples

- AI-powered drones
- Automated missile systems

Ethical Concerns

- Lack of human control in life-and-death decisions
- Increased risk of unintended harm
- Potential misuse in warfare

Impact

- Escalation of conflicts
- Civilian casualties
- Global security risks

Possible Solutions

- International regulations and treaties
- Human-in-the-loop control systems
- Ethical guidelines for military AI

4. Lack of Accountability

What is Accountability in AI?

Determining who is responsible when an AI system causes harm or makes a wrong decision.

Key Issues

- Is it the developer, company, or user?
- AI systems can be complex and hard to understand (black-box problem)

Examples

- Self-driving car accidents
- Incorrect medical diagnosis by AI

Impact

- Legal challenges

- Difficulty assigning blame
- Reduced trust in AI systems

Possible Solutions

- Clear legal frameworks
- Explainable AI (transparent decision-making)
- Proper testing and validation